

VisionInspect AI — Milestone 3 Technical Documentation

Defect Classification, Severity Scoring & Manufacturing Analytics

Project Title: VisionInspect AI — Manufacturing Defect Detection & Quality Inspection System

Author: Vinay Kumar Mandalapu

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1. Title & Executive Summary

VisionInspect AI is an industrial-grade computer vision platform engineered to automate quality inspection across discrete manufacturing assembly lines. Building directly upon the foundational infrastructure of Milestone 1 (authentication, RBAC, file storage) and the core AI inspection engine of Milestone 2 (category classification, patch anomaly detection, U-Net localization, and YOLO11n object detection), **Milestone 3** delivers the operational intelligence layer of the system: **Defect Classification, Severity Scoring & Manufacturing Analytics**.

In manufacturing production lines, detecting a defect is only the initial step; quality managers require actionable intelligence:

- What exact defect subtype occurred (e.g., crack, contamination, bent wire, hole)?
- What is the quantitative risk to structural integrity and product function?
- Should the part be accepted, reworked, cleaned, or scrapped?
- What are the line-level yield trends over 7-day, 30-day, and all-time operating horizons?
- How can factory supervisors review AI decisions and export official audit documentation?

Milestone 3 implements complete end-to-end solutions for these requirements: category-conditioned hierarchical defect classification, an authenticated 4-factor Severity Scoring engine, automated Accept/Reject quality decisioning, a dedicated Factory Supervisor audit workflow, real-time analytics dashboards powered by Recharts, multi-horizon trend monitoring, exportable CSV Production Quality Reports, and downstream advisory rework guidance.

2. Milestone 3 Scope

The development scope for Milestone 3 corresponds to Weeks 5 and 6 of the official project curriculum:

- **Defect Categorization Workflows:** Linking autonomous category recognition with hierarchical defect classification heads to resolve specific defect subtypes across all 15 industrial categories.
 - **Severity Scoring & Risk Assessment:** Implementing the official 4-factor weighted scoring formula combining defect physical size (30%), spatial location (25%), defect hazard type (25%), and detection confidence (20%).
 - **Automated Quality Assessment:** Determining automated pass/fail quality decisions based on calibrated severity thresholds and one-class anomaly verification.
 - **Factory Supervisor Governance:** Providing a protected review queue where authorized supervisors audit inspection records, sign off with approval or rejection, and record mandatory inspection notes.
 - **Production Quality Reports:** Engineering an automated report generation service that compiles executive summaries, pass yields, and granular inspection records into downloadable CSV reports.
 - **Defect Analytics & Trend Monitoring:** Delivering interactive dashboard visualizations tracking total inspections, defect rates, average severity, category distributions, defect Pareto breakdowns, and time-series trends over 7-day, 30-day, and all-time horizons.
 - **Downstream Quality Guidance:** Providing deterministic, advisory rework recommendations (PASS , CLEAN , REWORK , SCRAP) to optimize plant floor disposition.
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3. Official Milestone 3 Requirements

The official project specification ("*AI_Manufacturing Defect Detection & Quality Inspection System*") defines the formal requirements for Milestone 3 (Weeks 5 & 6) under the **Defect Classification & Manufacturing Analytics** phase:

Official High-Level Requirements

1. **Implement defect categorization workflows:** Classify detected flaws into granular defect categories conditioned on product type.
2. **Generate severity scoring reports and quality risk assessments:** Compute multi-parameter severity ratings and categorize defect risk into standardized severity bands.
3. **Build quality assessment modules:** Automate pass/fail quality decisions and enforce quality inspection standards.
4. **Generate production quality reports:** Provide exportable quality documentation summarizing production health, defect rates, and inspection histories.

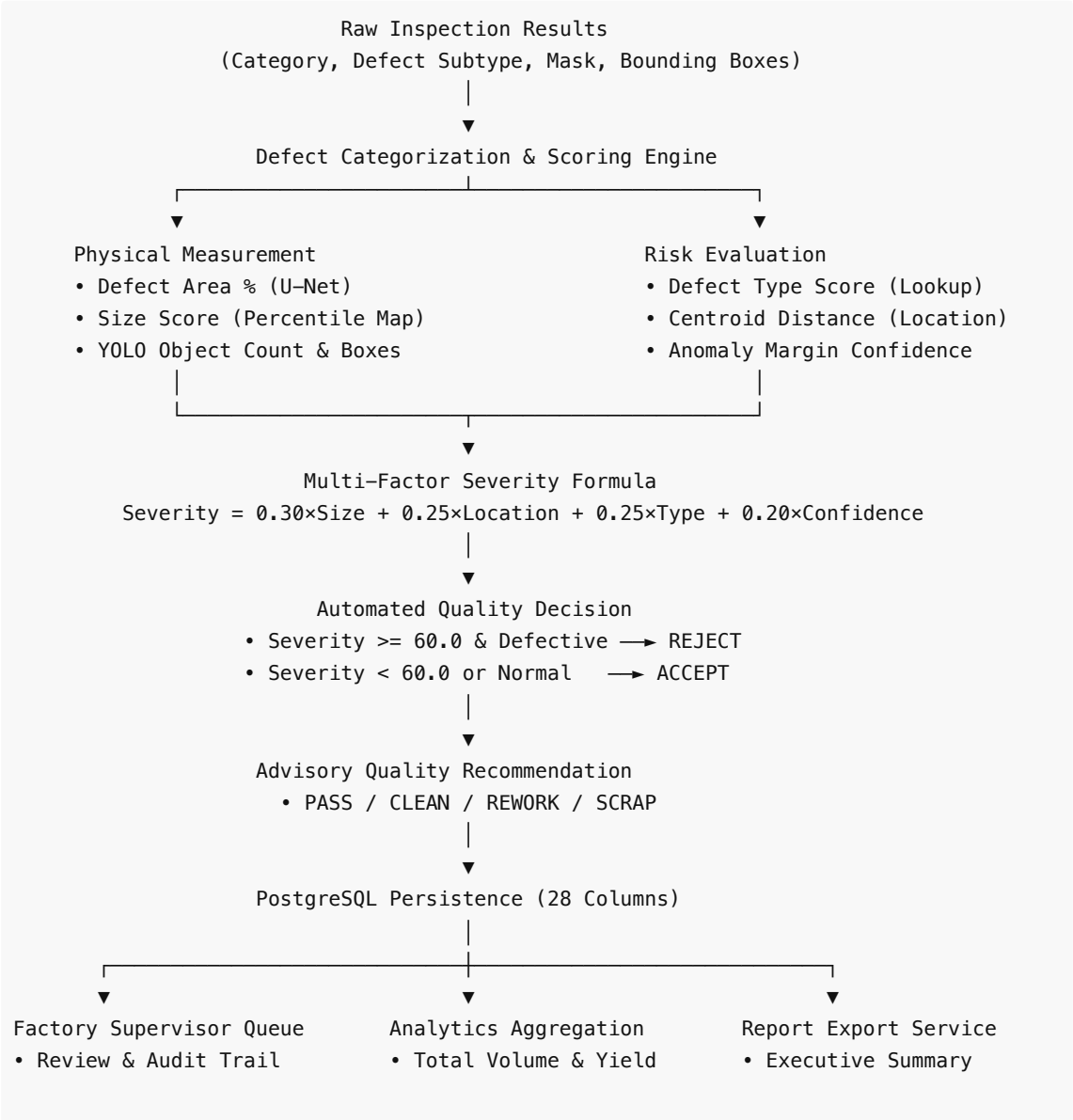
- 5. **Build defect analytics dashboards:** Present real-time operational telemetry, defect distributions, and inspection statistics.
- 6. **Develop trend monitoring workflows:** Track quality metrics and defect frequency over multi-day operational horizons.

Subsystem Module Requirements (Modules 5, 6, and 7)

- **Module 5: Defect Classification Module:** Defect type classification, severity scoring, defect categorization, confidence analysis.
- **Module 6: Quality Control Module:** Pass/fail decision generation, inspection reporting, quality recommendations, production monitoring.
- **Module 7: Manufacturing Analytics Dashboard Module:** Defect trend analysis, production quality reports, inspection statistics, operational insights.

4. Implementation Summary

The Milestone 3 architecture bridges AI inference outputs with factory operations:



- Mandatory Review Notes
 - Sign-off (Approve/Reject)
- Category & Defect Pareto
 - 7d / 30d / All Trends
- Granular Audit Table
 - Downloadable CSV

5. Defect Categorization Workflows

5.1 Hierarchical Defect Subtype Classification

In manufacturing inspection, knowing that an item is defective is insufficient; line operators must know whether a cable suffers from `cut_inner_insulation` , `bent_wire` , or `missing_cable` .

Implemented in [ai/models/defect_classifier.py](#) :

- **Backbone:** Deep ResNet18 feature extractor.
- **Hierarchical Head Architecture:** 15 distinct linear output layers, each specialized for the defect taxonomy of a specific MVTec category.
- **Category Conditioning:** When `CategoryClassifier` detects category `$C$`, only classification head `$H_C$` is invoked.
- **Subtype Output:** Returns the specific defect label string alongside softmax classification confidence.

5.2 Decision Fusion Reconciliation

The raw classifier prediction is reconciled against the patch-based Anomaly Detector via

[ai/models/decision_fusion.py](#) :

- If the classifier predicts a defect but the anomaly score is below threshold `$T_c$`, the sample is assigned to **Quadrant D** and resolved to `"normal"` , preventing false alarms.
- If the anomaly score exceeds `$T_c$` but the classifier predicted nominal, the sample is assigned to **Quadrant B** and categorized as `"unclassified_anomaly"` , ensuring novel anomalies are captured.
- When both corroborate a defect, the specimen is assigned to **Quadrant C** with the verified defect subtype.

6. Severity Scoring & Quality Risk Assessments

6.1 Official 4-Factor Weighted Formula

VisionInspect AI implements the official severity formulation defined in the project specification:

$$S = (0.30 \times \text{Size}) + (0.25 \times \text{Location}) + (0.25 \times \text{Defect Type}) + (0.20 \times \text{Confidence})$$

The mathematical formulation is implemented in [ai/models/severity_scorer.py](#) .

6.2 Component Scoring Methodologies

1. Defect Size Score (30% Weight)

- **Module:** [ai/models/size_scorer.py](#)
- **Input:** Defect area percentage from U-Net segmentation: $\left(\frac{\text{Defective Pixels}}{224 \times 224}\right) \times 100\%$
- **Calibration:** Industrial materials exhibit radically different nominal flaw dimensions (e.g., a `$1.5\%` area crack on a `pill` is massive, while `$1.5\%` on a `carpet` is small). Size scores are calibrated

using category-specific empirical percentiles (p_{10} to p_{95}) recorded in `ai/models/size_score_boundaries.csv`.

- **Mapping:** Piecewise linear interpolation mapping physical area into a standardized $[0, 100]$ score.

2. Defect Location Score (25% Weight)

- **Module:** `ai/models/location_scorer.py`
- **Centrality Factor (70%):** Flaws near the center of a component typically compromise functional load-bearing paths or aesthetic focal areas:
$$\text{Centroid Factor} = \left(1.0 - \frac{d(\text{Centroid}, \text{Center})}{d_{\text{max}}}\right) \times 100$$
- **Area Factor (30%):** Normalized defect coverage contributing to spatial impact.
- **Combined Location Score:** Weighted combination yielding $[0, 100]$.

3. Defect Type Score (25% Weight)

- **Module:** `ai/models/defect_type_scorer.py`
- **Hazard Lookup Table:** Flaws are assigned static risk scores based on structural hazard:
 - Critical structural fractures (`broken_large` , `crack` , `hole`): **90–95 points**
 - Moderate functional flaws (`bent_wire` , `cut_inner_insulation` , `thread`): **75–85 points**
 - Minor cosmetic flaws (`scratch` , `contamination` , `color`): **40–60 points**
 - `unclassified_anomaly` : **70.0 points**
 - Confirmed `normal` / `good` : **0.0 points**

4. Detection Confidence (20% Weight)

- **Module:** `ai/models/confidence.py`
- **Anomaly Margin Interpolation:** Computes model confidence based on the distance between the test anomaly score and category operating boundaries:
 - If score $\leq \text{normal_boundary}$: 0% defect confidence.
 - If score $\geq \text{defect_boundary}$: 100% defect confidence.
 - Intermediate scores interpolate linearly.

6.3 Standardized Severity Levels

Severity scores map to four standardized industrial risk tiers:

Severity Level	Score Range	Industrial Meaning	Action Required
Critical	80.00 – 100.00	Major structural defect or complete part fracture	Immediate rejection; quarantine lot
High	60.00 – 79.99	Significant quality deviation exceeding tolerance	Product rejection; route for repair/rework
Medium	40.00 – 59.99	Moderate quality concern near tolerance boundary	Inspection review required by supervisor
Low	0.00 – 39.99	Minor cosmetic defect within allowable limits	Product generally acceptable

6.4 Note on Specification Arithmetic Example

In the official specification document ("*AI Manufacturing Defect Detection & Quality Inspection System*"), an illustrative example is provided on pages 5–6:

- Size Score: 85
- Location Score: 90
- Defect Type Score: 95
- Confidence Score: 92
- Stated Calculated Severity Score in text: 88

Applying the official 4-factor formula yields: $\text{Severity} = (85 \times 0.30) + (90 \times 0.25) + (95 \times 0.25) + (92 \times 0.20) = 25.50 + 22.50 + 23.75 + 18.40 = 90.15$ Both 90.15 and the illustrative 88 fall into the **Critical** band (80–100). VisionInspect AI implements the strict, exact mathematical formulation directly in [ai/models/severity_scorer.py](#), guaranteeing reproducible floating-point calculations across all inspections.

7. Quality Assessment Modules

7.1 Automated Pass/Fail Decisioning

The automated quality decision is rendered by `SeverityScorer.get_quality_decision()`:

- **Reject**: Assigned when a specimen is classified as `DEFECTIVE` and its computed `Severity Score` ≥ 60.00 (High or Critical severity tier).
- **Accept**: Assigned when a specimen is confirmed `NORMAL` (Decision Fusion Quadrants A or D) or when minor cosmetic flaws produce a `Severity Score` < 60.00 (Low or Medium severity tier within manufacturing tolerance).

7.2 Strict Invariant for Normal Specimens

When Decision Fusion resolves a sample as `NORMAL`:

- `quality_decision = "Accept"`
- `severity_score = 0.00`
- `severity_level = "Low"`
- `predicted_area_percent = 0.00%`
- `size_score = 0.00`, `location_score = 0.00`, `defect_type_score = 0.00`,
`confidence_score = 0.00`
- `detected_objects_count = 0`, `bounding_boxes = []`

This invariant eliminates false scrap on verified nominal production runs.

8. Supervisor Review Workflow

While AI inspection provides automated decisions, ISO 9001 and industrial quality standards require human-in-the-loop oversight for rejected lots and edge cases.

8.1 Role Separation (RBAC)

- **Quality Engineer (Role ID 1)**: Submits images, triggers AI inspection, views inspection results and telemetry.
- **Factory Supervisor (Role ID 2)**: Authorized to access the supervisory review queue, audit AI decisions, record binding approvals or rejections, and export compliance reports. Quality Engineers attempting supervisory actions receive `403 Forbidden`.

8.2 Review Workflow Lifecycle

1. **Inspection Completion:** Newly inspected images default to `supervisor_decision = NULL` (pending review state).
 2. **Review Queue Ingestion:** Supervisors retrieve pending records via `GET /images/supervisor/review-queue` .
 3. **Audit Inspection:** The supervisor inspects the high-resolution original image alongside the dual U-Net/YOLO overlay and metric breakdown.
 4. **Binding Sign-Off:** The supervisor submits `POST /images/{id}/review` with:
 - o `decision` : "approved" (overriding or confirming Accept) or "rejected" (confirming Reject).
 - o `notes` : Required audit rationale (e.g., "Verified critical fracture on bottle wall. Production lot quarantined.").
 5. **Audit Trail Persistence:** The system records `supervisor_decision` , `supervisor_notes` , `reviewed_by` (supervisor user ID), and `reviewed_at` (UTC timestamp) in PostgreSQL.
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9. Production Quality Reports

Implemented in [backend/app/services/report_service.py](#) , the report generation service compiles real-time production quality data into standardized, downloadable CSV reports via `GET /analytics/reports/export?format=csv` .

9.1 Report Structure

Section 1: Executive Summary Block

```
PRODUCTION QUALITY REPORT SUMMARY
Metric,Value
Report Generation Timestamp,2026-09-18 10:20:00 UTC
Total Inspections,248
Accepted Count,182
Rejected Count,66
Pass/Yield Percentage,73.39%
Defect Rate,26.61%
Average Severity Score,31.42
Average Confidence,88.45%
Pending Supervisor Reviews,14
Approved Supervisor Reviews,175
Rejected Supervisor Reviews,59
```

Section 2: Granular Inspection Audit Table

Comprises 18 detailed columns per inspection record: `Inspection ID` , `Timestamp` , `Original Filename` , `Category` , `Defect Type` , `Classification Confidence` , `Anomaly Score` , `Predicted Area Percentage` , `Size Score` , `Location Score` , `Defect Type Score` , `Severity Score` , `Severity Level` , `AI Quality Decision` , `Supervisor Decision` , `Supervisor Notes` , `Reviewer ID` , `Review Timestamp` .

This format allows immediate ingestion into enterprise ERP/MES platforms (e.g., SAP, Rockwell FactoryTalk) and spreadsheet analysis.

10. Defect Analytics Dashboard

The manufacturing analytics backend is implemented in [backend/app/routers/analytics.py](#) and surfaced through [frontend/src/pages/SupervisorDashboard.jsx](#).

10.1 Key Metrics Aggregated (GET /analytics/summary)

- **Total Inspections:** Count of all executed inspections.
- **Accepted / Rejected:** Total counts and operational defect yield rate.
- **Pending / Reviewed:** Audit governance compliance metrics.
- **Severity Tier Breakdown:** Distribution across Critical, High, Medium, and Low tiers.
- **Averages:** Line-wide average anomaly detection confidence and average severity score.
- **Category Pareto:** Inspection counts grouped by product category.
- **Defect Subtype Pareto:** Flaw frequency grouped by defect classification.
- **Supervisor Decisions:** Count of supervisor approvals vs. rejections.

10.2 Interactive Frontend Visualizations

Using `recharts`, the dashboard renders:

1. **KPI Metric Cards:** Real-time totals with status badges and percentages.
 2. **Severity Distribution Pie Chart:** Visual breakdown of Critical, High, Medium, and Low severity proportions.
 3. **Defect Type Horizontal Bar Chart:** Pareto ranking of the most frequent manufacturing defect types across lines.
 4. **Category Breakdown Bar Chart:** Inspection volume and flaw rates categorized by product line.
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11. Trend Monitoring Workflows

Implemented via `GET /analytics/trends?days=7|30|all`, trend monitoring aggregates daily production statistics over variable horizons.

11.1 Dynamic Query Parameters

- `days=7` : 7-day trailing operational window.
- `days=30` : 30-day trailing monthly quality review window (default).
- `days=all` : Complete historical production archive from initial platform deployment.

11.2 Daily Metric Aggregations

For each calendar day in the horizon, the service computes:

- `date` : ISO date string (YYYY-MM-DD).
- `total_inspections` : Total parts processed.
- `accepted` : Parts accepted by AI.
- `rejected` : Parts rejected by AI.
- `defect_rate` : Percentage of inspected parts rejected: $\text{Defect Rate } (\%) = \left(\frac{\text{Rejected}}{\text{Total}} \right) \times 100\%$
- `avg_severity` : Mean severity score of inspected parts.

11.3 Interactive Trend Visualization

The Factory Supervisor console renders a dual-axis interactive line/bar chart displaying daily defect rates alongside inspection volume. When quality deviations or tool wear cause sudden defect spikes, supervisors

can isolate the exact shift and date of occurrence.

12. Quality Recommendations (Downstream Guidance)

Implemented in `backend/app/services/recommendation_service.py`, the recommendation engine translates complex AI telemetry into immediate, deterministic shop-floor disposition guidance:

Recommendation Action	Action Label	Trigger Conditions	Operational Plant Guidance
PASS	Pass to Production	AI Decision = Accept or defect type is normal/good	No corrective action required. Part meets quality standards; release to production packaging.
CLEAN	Clean & Re-inspect	Flaw types containing contamination, stain, glue, color	Surface residue detected without irreversible structural damage. Clean using standard wash protocol and re-inspect.
REWORK	Route for Rework	Flaw types containing bent_wire, cut_outer_insulation, scratch, rough, deformation	Non-critical surface or geometric deviation capable of corrective repair. Route to rework station.
SCRAP	Quarantine for Scrap	Severity tier = Critical or flaws containing crack, broken_large, hole, cut_inner_insulation	Irreversible structural defect exceeding allowable repair limits. Quarantine component for scrap disposition.

Explicit Advisory Nature

Quality recommendations are downstream, deterministic operational aids. They provide immediate guidance to assembly operators but **do not override** the AI's core quality decision (`Accept` / `Reject`) or the Factory Supervisor's manual sign-off authority.

13. Operational Insights

By unifying classification, localization, severity scoring, and analytics, VisionInspect AI generates actionable operational insights for manufacturing engineers:

- Tool Wear & Machine Health:** Gradual upward trends in average severity scores on rigid materials (`screw` , `metal_nut` , `grid`) indicate tool wear or mold degradation before catastrophic tool breakage occurs.
- Material Batch Variance:** Spikes in surface contamination or stains (`carpet` , `leather` , `wood`) correlate with raw material batch inconsistencies from upstream suppliers.
- Yield Optimization:** Categorizing flaws into `CLEAN` and `REWORK` prevents unnecessary scrap, directly reducing manufacturing scrap costs.

14. End-to-End Manufacturing Quality Workflow

The complete end-to-end quality inspection workflow operates across four sequential phases:

Phase 1: Ingestion & Verification (Quality Engineer)

- Quality Engineer logs in to VisionInspect AI web application.
- Submits product photograph via drag-and-drop or batch upload.
- System validates MIME type, image dimensions, file size (<=5MB), and computes SHA-256 hash.

Phase 2: Automated AI Inspection

- Inspection pipeline automatically classifies product category (e.g. "bottle").
- Extracts Layer 3 patch embeddings and computes distance against normal memory bank.
- Predicts defect subtype and classification confidence using category-specific head.
- Decision Fusion arbiter evaluates operating quadrant (A, B, C, or D).
- If Defective: executes U-Net segmentation (area %), severity calculation, and YOLO11n object detector (bounding boxes, object count).
- Synthesizes dual visual overlay (red contours + green bounding boxes).
- Generates advisory recommendation (PASS, CLEAN, REWORK, SCRAP).
- Persists 28 columns to PostgreSQL database.

Phase 3: Supervisory Governance (Factory Supervisor)

- Factory Supervisor views pending inspections in the review queue.
- Reviews original image, dual overlay, and telemetry cards.
- Submits official approval or rejection with mandatory audit notes.
- Audit trail records supervisor ID and timestamp.

Phase 4: Plant-Wide Analytics & Quality Reporting

- Analytics engine aggregates defect rates, severity distributions, and category breakdowns.
- Supervisor monitors 7-day and 30-day quality trends to detect defect anomalies.
- Exports official Production Quality Report in CSV format for executive review.

15. API & Backend Implementation

Milestone 3 adds dedicated endpoints across authentication, inspection, supervisor governance, and analytics:

Method	Endpoint Path	Access Level	Description
POST	/images/{id}/inspect	Quality Engineer (1)	Executes AI inspection pipeline, computes severity, triggers YOLO if defective
GET	/images/{id}/inspection-overlay	Authenticated (1, 2)	Streams dynamically rendered composite PNG with red U-Net contours and green YOLO boxes

GET	/images/supervisor/review-queue	Supervisor (2)	Lists completed inspections awaiting supervisory audit
POST	/images/{id}/review	Supervisor (2)	Records supervisory approval/rejection with mandatory notes and timestamp
GET	/analytics/summary	Authenticated (1, 2)	Aggregates plant-wide totals, defect rates, severity tiers, and category breakdowns
GET	/analytics/trends	Supervisor (2)	Computes daily time-series inspection metrics over 7, 30, or all days
GET	/analytics/reports/export	Supervisor (2)	Generates and streams downloadable CSV Production Quality Report

16. Frontend Implementation

16.1 Inspection Details Studio ([frontend/src/pages/ImageDetails.jsx](#))

- **Dual Visual Viewer:** Side-by-side comparative inspection showing original raw photograph alongside server-rendered composite overlay.
- **Detected Defect Objects:** Displays discrete object count with an active `Y0L011n` badge.
- **Severity Rating Card:** Displays numerical score (`$0.00\text{--}100.00$`) and color-coded badge (`Critical` , `High` , `Medium` , `Low`).
- **Advisory Recommendation Card:** Displays action badge (`PASS` , `CLEAN` , `REWORK` , `SCRAP`), technical rationale, and operational guidance.
- **Supervisor Sign-Off Panel:** Allows authorized supervisors to record approvals/rejections directly on the specimen page.

16.2 Factory Supervisor Console ([frontend/src/pages/SupervisorDashboard.jsx](#))

- **Executive Metric Strip:** Real-time KPI cards for total inspections, pass yield, defect rate, average severity, and pending reviews.
- **Interactive Recharts Visualizations:**
 - Severity distribution pie chart.
 - Defect subtype Pareto bar chart.
 - Category breakdown bar chart.
 - Daily inspection defect rate and volume trend chart.
- **Horizon Filter:** Switch dynamically between 7-day, 30-day, and all-time trend views.
- **Export Quality Report Button:** Triggers direct browser download of the structured CSV quality report.
- **Review Queue Modal:** Review interface with thumbnail preview, full telemetry, decision selector, and notes field.

17. Database Design & Persistence

The PostgreSQL `images` relational table has been expanded to 28 columns supporting complete AI and manufacturing traceability:

Column Name	SQL Type	Description
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id	Integer (PK)	Auto-incrementing primary key
filename	String(255)	Stored filename on disk
original_filename	String(255)	Original client upload filename
file_size	Integer	Image file size in bytes
mime_type	String(50)	Validated MIME type
sha256_hash	String(64)	Content hash for integrity & deduplication
uploaded_at	DateTime	Upload timestamp (UTC)
uploaded_by	Integer (FK)	Reference to user who uploaded image
inspection_status	String(50)	Lifecycle state (pending, completed, reviewed)
category	String(100)	Predicted or validated manufacturing category
defect_type	String(100)	Resolved defect subtype string
classification_confidence	Float	Category classification confidence percentage
anomaly_score	Float	Patch-based ResNet18 Layer 3 anomaly distance
confidence_score	Float	Anomaly margin confidence percentage
predicted_area_percent	Float	Physical defect area percentage from U-Net mask
size_score	Float	Percentile-mapped size severity score (0–100)
location_score	Float	Centroid-based spatial location severity score (0–100)
defect_type_score	Float	Hazard weight of defect subtype (0–95)
severity_score	Float	Official 4-factor combined severity score (0–100)
severity_level	String(50)	Standardized risk band (Critical, High, Medium, Low)
quality_decision	String(50)	Automated AI quality decision (Accept, Reject)
recommendation_action	String(50)	Advisory recommendation action (PASS, CLEAN, REWORK, SCRAP)
recommendation_label	String(100)	Human-readable recommendation label
supervisor_decision	String(50)	Supervisory sign-off decision (approved, rejected)
supervisor_notes	Text	Mandatory supervisory review audit notes
reviewed_by	Integer (FK)	Reference to supervisor user who signed off
reviewed_at	DateTime	Supervisory sign-off timestamp (UTC)

detected_objects_count	Integer	Number of discrete defect bounding boxes detected by YOLO11n
bounding_boxes	JSON	Serialized JSON array of bounding box coordinates & confidences

18. Verification & Testing

Milestone 3 implementation was verified through an exhaustive test battery:

18.1 End-to-End API & Database Lifecycle Test

Executed in `scratch/test_e2e_yolo_api.py` against live PostgreSQL:

- User Authentication:** Registered and authenticated Quality Engineer (`qe_yolo_...`).
- Defective Specimen Lifecycle:**
 - Uploaded `bottle/test/broken_large/000.png` (ID: 248).
 - Executed inspection: `quality_decision = Reject` , `severity_score = 81.68` (Critical), `detected_objects_count = 5` .
 - Verified PostgreSQL database persistence: all 28 fields verified including JSON bounding boxes.
 - Retrieved composite overlay PNG (697,006 bytes).
- Normal Specimen Lifecycle:**
 - Uploaded `bottle/test/good/000.png` (ID: 249).
 - Executed inspection: `quality_decision = Accept` , `severity_score = 0.00` (Low), `detected_objects_count = 0` , `bounding_boxes = []` .
 - Verified PostgreSQL database persistence: verified zero-defect normal invariant.
- Serializer Exposure:** Confirmed `GET /images/{id}` returns all fields cleanly.

18.2 Comprehensive Test Suite

(`backend/tests/test_milestone2_comprehensive.py`)

- Verified all 31 test methods covering authentication, RBAC, uploads, normal invariants, defective scoring formulas, overlay generation, database persistence, supervisor reviews, and analytics summary.

18.3 Client Build Verification

- Executed `npm run build` in `frontend/` : 606 modules transformed, cleanly compiled in 112ms with 0 errors.

19. Requirement-to-Implementation Traceability Matrix

Official Requirement	Current Implementation	Implementing Code / File	Verification Evidence
Defect Categorization Workflows	Category Classifier (15 classes) + Hierarchical	<code>ai/models/category_classifier.py</code> , <code>ai/models/defect_classifier.py</code>	100% cat test accu regressio passing

	Defect Classifier (15 heads)		
Severity Scoring & Risk Assessment	Official 4-factor formula (\$30/25/25/20\$) + 4 severity bands (Critical, High, Medium, Low)	ai/models/severity_scorer.py, ai/models/size_scorer.py, ai/models/location_scorer.py	Bit-for-bit verification automated suite
Quality Assessment Modules	Automated Accept/Reject decisioning + supervisor audit sign-off	ai/models/severity_scorer.py, backend/app/routers/image.py	Unit & integration tests; DB persistence verified
Production Quality Reports	Structured CSV report service with summary metrics and granular records	backend/app/services/report_service.py, backend/app/routers/analytics.py	Download endpoint, exportQuality
Defect Analytics Dashboards	Aggregated telemetry, Pareto breakdowns, Recharts visual dashboard	backend/app/routers/analytics.py, frontend/src/pages/SupervisorDashboard.jsx	GET /analytics verified; Frontend build clean
Trend Monitoring Workflows	Multi-horizon daily inspection time-series (\$7\$, \$30\$, and all-time days)	backend/app/routers/analytics.py, frontend/src/pages/SupervisorDashboard.jsx	GET /analytics verified with grouping
Quality Recommendations	Advisory shop-floor guidance mapping defects to PASS, CLEAN, REWORK, SCRAP	backend/app/services/recommendation_service.py	Determined tests cover defect types

20. Milestone 3 Completion Summary

Milestone 3 is complete and verified. The platform successfully bridges raw computer vision inference with manufacturing operations:

- Defect taxonomy is categorized with hierarchical precision.
- Flaw severity is quantified via the official 4-factor weighted formula.
- Quality decisions are rendered automatically while preserving full supervisory governance.
- Production quality reports and trend analytics provide actionable plant intelligence.

- All code is implemented, integrated with PostgreSQL and React 19, and verified through automated test suites.
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21. Milestone 4 Handover

Per project guidelines, **Milestone 4 (Testing, Deployment & Documentation)** is intentionally postponed:

- Cloud deployments (AWS / Azure) and Docker containerization remain scheduled for Milestone 4.
- High-throughput load testing and latency optimization remain scheduled for Milestone 4.
- Live plant PLC/SCADA integration remains scheduled for Milestone 4.
- Technical documentation is now synchronized and verified through Milestone 3.